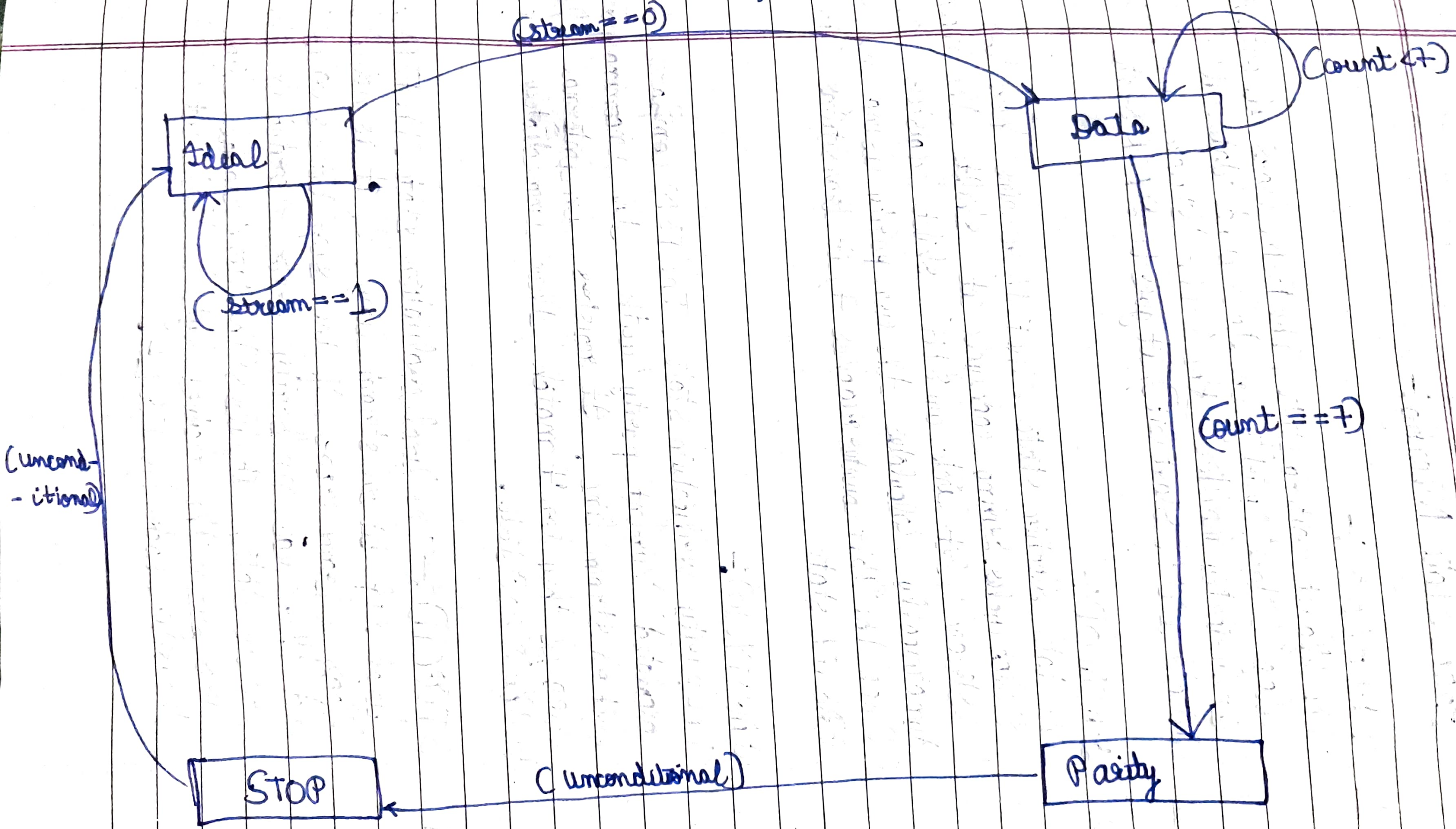
State diagram



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Each State Meaning:-

1 IDLE(00):-
The default waiting state. The FSM clears all output flags ('done', 'parity_err etc) and 'data_out' to zero. It constantly monitors the 1-bit 'stream' line. If it detects a '0' (Start Bit), it goes to DATA State

2 DATA(01):-
The data collection state. The FSM remains here for 8 clock cycles. During each cycle, it shifts the incoming 'stream' bit into an 8 bit shift register and dynamically calculates the even parity using an XOR operation. Once the 4bit counter reaches 7, it transitions into PARITY state

3 PARITY(10):-
The parity verification state. The FSM compares final calculated 'current_parity' with the incoming parity bit on stream' line. It registers an internal parity flag if they don't match, then immediately transitions to the STOP state.

4 STOP(11):- The final evaluation state. The FSM checks final stop bit on 'stream' line. If it is not a '1', it flags a frame error. If the stop bit is valid and there was no parity error recorded in the previous state, it pulses the 'done' signal high and pushes the recieved byte to 'data_out'. It then transitions back to IDLE.